Correlation and Regression

Bivariate distribution

The distribution in which we considered two variables simultaneously for each item of the series, distribution of height and wt. of a person.

- The change in one variable effects a change in other variable. The variable are said correlated

→ Mutual linear inter dependance of two or more variable is called **correlation**.

- If changes, some direction → +ve Co..
- " " → opposite direction → -ve Co...

The measure of correlation called the co-efficient (denoted by r) → measure the closeness of the relationship between two variable.

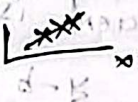
Method of studying co-relation

a) Scatter diagram method.
b) Karl pearson's co-efficient of correlation.
c) Spearman's Rank correlation co-efficient.
d) Method of least squares.

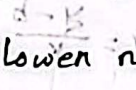
*) **Scatter diagram method**: → (ploted on a graph paper) (x_i, y_i) → by looking.

- ✓ more scatter to the line → less relation → less correlated
- ✓ less scatter on close to the line → more correlated

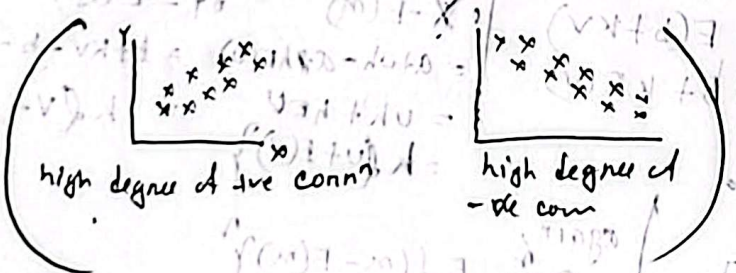
1) If the point lie on the straight line from (left hand corner → upper right or) is is perfectly correlated. $(r = +1)$



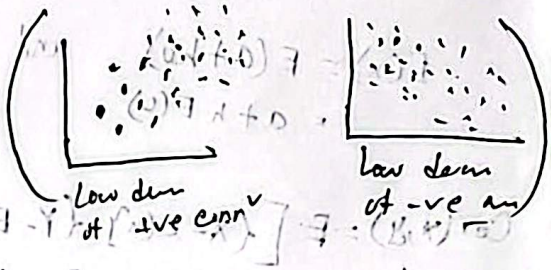
2) straight line follow but upper left hand corner to lower right hand corner, corr^n is said perfectly negative. $(r = -1)$



3) narrow band



widely scattered



4)



haphazardly manner.

No corr^n $(r = 0)$

B. Karl pearson's correlation coefficient: Unit free correlation

1920

$$r_{xy} = \frac{Cov(x,y)}{\sqrt{Var(x).Var(y)}} = \frac{\frac{1}{n}\Sigma(x_i-\bar{x}).(y_i-\bar{y})}{\sqrt{\left\{\frac{1}{n}\Sigma(x_i-\bar{x})^2\right\}\left\{\frac{1}{n}\Sigma(y_i-\bar{y})^2\right\}}}$$

Working formula $= \dfrac{\Sigma x_i y_i - \frac{(\Sigma x_i)(\Sigma y_i)}{n}}{\sqrt{\left\{\Sigma x_i^2 - \frac{(\Sigma x_i)^2}{n}\right\}\left\{\Sigma y_i^2 - \frac{(\Sigma y_i)^2}{n}\right\}}}$

Assumption:

1) Both variable are measured in nominal interval and ratioscale.
2) The two variable follow bi-variate normal distribution.
3) The relationship between the variable is linear
4) The sample are in adequate size to assure normality.

misinterpretation: r is a measure of linear association. If $r=0$ that does not mean they are not correlated, you can say they are not linearly correlated.

Properties:

1) The pearson's correlation co-efficient lies -1 to 1. If r is zero they are uncorrelated and also independent variable are a zero correlation coefficient
2) The correlation coefficient measures the amount of change and also the direction of change.
3) It is based on all observation
4) It measure the amount of linear relation.
5) It is affected by large, smaller or extreme values.
6) The correlation co-efficient is independent of change and origin

proof:

Let $U = \frac{x-a}{h}$, $V = \frac{y-b}{k}$ so, $x = uh+a$ and $y = vk+b$

Here a, h, b, k are constant, Now need to prove $r(x,y) = r(U,V)$

$E(x) = E(a+hu)$
$= a + hE(u)$

and

$E(y) = E(b+kV)$
$= b + kE(V)$

$x - E(x) = a+uh - a - hE(u) = uh + hEu = h\{u + E(u)\}$

$y - E(y) = b+kv - b - kE(v) = k\{v - E(v)\}$

$Cov(x,y) = E[\{x-E(x)\}\{Y-E(Y)\}]$
$= hk\, E[\{u+E(u)\}.\{v+E(v)\}]$
$= hk\, cov(U,V)$

again,
$\sigma_x^2 = E\{(x-E(x)\}^2$
$= h^2 E\{u-E(u)\}^2$
$= h^2 \sigma_u^2$

$\boxed{\sigma_x = h\sigma_u}$ and $\boxed{\sigma_y = k\sigma_v}$

Now, $C(X,Y) = \frac{cov(X,Y)}{\sigma_X \cdot \sigma_Y} = \frac{h.k\ cov(U,V)}{h\sigma_U, k\sigma_V}$

$= \frac{cov(U,V)}{\sigma_U \sigma_V}$

hence prove, independent of scale and chn.

Regression

Regression analysis is a mathmatical measure of the average relationship between two or more variable, expressed in a original unit of the data.

Types of variable:

a) Dependent / Regressed / explained variable → Value which is influenced on predicted.
b) Independent / Regresson / predictor / explanatory variable → influce the value which is used for prediction.

Uses:

a) estimate the relationship between depended and explanatory variable.
b) Predict the dependent variable.

Simple linear Regression Model:

For random sample (X_i, Y_i), $i = 1, 2, \ldots n$ from a bivariate distribution with Y dependent and x independent.

$$Y_i = a + bX_i + e_i$$

a = intercept term
b = slope
e = random error componets

assumption:

a) The x's are fixed
b) The regression of Y on x is linear
3) $e_i \sim NID(0, \sigma^2)$.
4) The e's are unconelated

Estimation by least square:

minimize the error sum of square.

$$\Sigma e_i^2 = \Sigma (Y - a - bX_i)^2$$

Normal equation: $\boxed{\hat{a} = \bar{Y} - b\bar{x}}$

Properties of Regression Co-efficient:

a) They are independent of change of origin but not of scale.

b) A regression co-efficient lies between $-\infty$ to $+\infty$

c) The two co-efficient cann't both exceed 1 at a same time. If $b_{yx} \geq 1$ then $b_{xy} \leq 1$ and vice versa.